

DDA5001 Machine Learning

Training versus Testing (Part III)

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Recap: Generalization for Finite Hypothesis Space

Theorem: Generalization for **finite hypothesis space**

Let $\mathcal{H}$ be a **finite** hypothesis space, i.e., $|\mathcal{H}| < \infty$. For any $\delta > 0$, the following generalization bound holds **with probability at least $1 - \delta$**

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{\log\left(\frac{2|\mathcal{H}|}{\delta}\right)}{2n}} \quad (1)$$

- ▶ More samples (larger n) lead to better generalization.
- ▶ The generalization error increase when $|\mathcal{H}|$ grows, but only **logarithmically**.
- ▶ However, it is only for finite hypothesis case, i.e., $|\mathcal{H}| < +\infty$. This is **impractical**.

Recap: Dichotomy, Growth Function, and VC Dimension

Dichotomies of $\mathcal{H}$

Given $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. The dichotomies generated by $\mathcal{H}$ on these points are defined by

$$\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_n) = \{(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)) : f \in \mathcal{H}\}.$$

Growth function

The growth function for the hypothesis set $\mathcal{H}$ is defined as:

$$\mathcal{G}_{\mathcal{H}}(n) = \max_{\{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathcal{X}} |\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_n)|$$

VC dimension

The VC dimension of a hypothesis space $\mathcal{H}$, denoted by $d_{\text{VC}}(\mathcal{H})$ or simply d_{VC} , is the largest n so that it can be shattered by $\mathcal{H}$, i.e.,

$$d_{\text{VC}}(\mathcal{H}) := \max\{n : \mathcal{G}_{\mathcal{H}}(n) = 2^n\}.$$

If $\mathcal{G}_{\mathcal{H}}(n) = 2^n$ for all n , then $d_{\text{VC}}(\mathcal{H}) = \infty$.

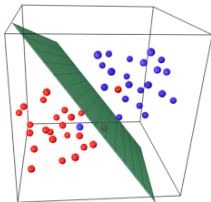
► These quantities measure the complexity of $\mathcal{H}$, even when $|\mathcal{H}| = \infty$.

Agenda

VC Dimension-induced Generalization

Example: VC Dimension of Linear Classifier

For a linear classifier, we can derive its VC dimension in a general sense.
This can be generalized to the following general result:



Theorem

For d -dimensional (binary) linear classifier, we have

$$d_{VC} = d + 1.$$

► Proof is put in the supplementary material.

Property of VC Dimension of Linear Classifier

- ▶ d_{VC} is exactly the number of parameters of a d -dimensional binary linear classifier (think about perceptron), i.e., $\theta_0, \theta_1, \dots, \theta_d$.
- ▶ d_{VC} measures the **effective number of parameters**, and hence the complexity of $\mathcal{H}$.
- ▶ The more parameters a model has, the more complex $\mathcal{H}$ is. This is reflected by a large d_{VC} .
- ▶ In some other models, the effective parameters may be less obvious.

VC Dimension Generalization Result

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After introducing all the related notions, we can now introduce the VC dimension generalization result.

VC generalization bound

For any $\delta > 0$, with probability at least $1 - \delta$, we have the following generalization bound:

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{8}{n} \log \left(\frac{4\mathcal{G}_{\mathcal{H}}(2n)}{\delta} \right)}$$

Upon invoking the upper bound on growth function using VC dimension, we have

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{8}{n} \log \left(\frac{4((2n)^{d_{\text{vc}}} + 1)}{\delta} \right)}$$

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► See the supplementary material for a proof sketch.

VC Generalization versus Previous Ones

- The VC generalization bound has the form

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \mathcal{O} \left(\sqrt{\frac{d_{\text{VC}}}{n}} \right)$$

where $\mathcal{O}$ is used to hide a $\sqrt{\log n / \delta}$ term and some constants.

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- ▶ Comparing the VC generalization bound to the finite $\mathcal{H}$ bound, it is easy to see that we not only replace $|\mathcal{H}|$ with $\mathcal{G}_{\mathcal{H}}$, but also change some constants. This is due to some technical issues. Fortunately, the overall idea is still maintained, that is, we use a much more reasonable effective number ($\mathcal{G}_{\mathcal{H}}$ or d_{VC}) to measure the complexity of $\mathcal{H}$ rather than using $|\mathcal{H}|$.

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- ▶ Larger n means that Er_{in} will generalize better to Er_{out} . When $n \rightarrow \infty$, we have $\text{Er}_{\text{in}} = \text{Er}_{\text{out}}$, which is consistent with our observation from the law of large numbers.

Is VC Generalization Bound Meaningful / Useful?

- ▶ The VC analysis is a **universal** result since it applies to all hypothesis space, learning algorithm, input space, probability distributions, binary targets (It can be extended to other target functions as well).
- ▶ Due to such a generality, **the bound is loose**.

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Though it is a little bit loose, it gives us useful guidance when conducting machine learning.

- ▶ It formally establishes the feasibility of learning for **infinite** $\mathcal{H}$. For $\mathcal{H}$ with finite d_{VC} , once we have enough training samples, learning is likely to be feasible.
- ▶ It tends to be **equally loose** for different models, enabling us to compare different models by comparing their d_{VC} . In real applications, model with smaller d_{VC} tend to generalize better than that with larger d_{VC} .
- ▶ It gives us some rules of thumb, e.g., about the number of training samples: **$n \geq 10 \times d_{VC}$** .

↪ We list several applications / guidance of the VC bound in practice.

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Suppose we want the result to hold with probability at least $1 - \delta$, and generalization error (error between Er_{in} and Er_{out}) to be less than some small number ε , we have

$$n \geq \frac{8}{\varepsilon^2} \log \left(\frac{4((2n)^{d_{\text{vc}}} + 1)}{\delta} \right).$$

Concisely, we need $n \geq \mathcal{O}\left(\frac{d_{\text{vc}} \log(1/\delta) \log n}{\varepsilon^2}\right)$.

Example: Estimating the Sample Complexity

Example:

Suppose that we have a learning model with $d_{VC} = 3$ and would like the generalization error to be at most 0.1 with confidence 90% (so $\varepsilon = 0.1, \delta = 0.1$). How big a data set do we need?

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$$n \geq \frac{8}{0.1^2} \log \left(\frac{4((2n)^3 + 1)}{0.1} \right).$$

Solving the above inequality gives $n \approx 22000$. $\square$

- ▶ This obtained sample complexity is much bigger than the previously said rule of thumb $n \geq 10 \times d_{VC}$, due to the fact that VC bound is quite loose.
- ▶ Nonetheless, the practical guidance is illustrated. With larger d_{VC} , we need more samples. This is consistent with practice.

Penalty for Model Complexity and Learning Curve

Example: Estimating the Er_{out}

In practice, we are often given $\mathcal{S}$. Hence, n is fixed. The question is what performance we can expect given n ?

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Suppose that $n = 10,000$ and we have a 90% confidence requirement ($\delta = 0.1$). What is the out-of-sample error can we guarantee with this confidence, given that $d_{\text{VC}} = 3$?

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By the generalization bound, we have

$$\begin{aligned}\text{Er}_{\text{out}}(f) &\leq \text{Er}_{\text{in}}(f) + \sqrt{\frac{8}{10000} \log \left(\frac{4((20000)^3 + 1)}{0.1} \right)} \\ &\approx \text{Er}_{\text{in}}(f) + 0.16.\end{aligned}\quad \square$$

The Fundamental Trade-off

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \mathcal{O} \left(\sqrt{\frac{d_{\text{VC}}}{n}} \right)$$

To make Er_{out} small:

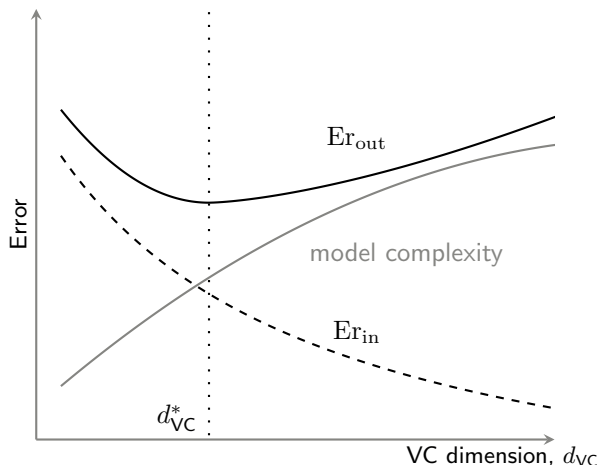
On the **training** side, we need:

more complex hypothesis $\mathcal{H}$ (larger d_{VC})

On the **generalization** side, we need:

less complex hypothesis $\mathcal{H}$ (smaller d_{VC})

Learning Curve from VC Analysis



- ▶ The **optimal model** is the one that minimizes the combinations of Er_{in} and generalization error.
- ▶ Occam's Razor principle: **The simplest workable model is the best.**

VC Generalization Result for Regression

VC Generalization Bound for Linear Regression

- ▶ So far our VC generalization bound is established for **binary classification** case where $y = \{-1, +1\}$.
- ▶ By adopting certain generalized notion like **pseudo-dimension**, we can apply the similar VC analysis to **linear regression model**, i.e., $y = \theta^\top x$ where y is **real-valued** (continuous). Such a generalization result then applies to linear regression.
- ▶ Similar to binary classification case, a d -dimensional linear regression model has **pseudo-dimension equal to $d + 1$** .

VC generalization bound for linear regression

$$\forall f \in \mathcal{H} \quad \text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \mathcal{O} \left(\sqrt{\frac{d_p}{n}} \right)$$

where d_p is the pseudo-dimension.

- ▶ See the book “Foundations of Machine Learning” Chapter 11.2 for details.

↪ Next lectures: Logistic regression and gradient-based optimization algorithm.